

Placement Empowerment Program
Cloud Computing and DevOps Centre

Set Up a Virtual Machine in the Cloud

Create a free-tier AWS, Azure, or GCP account. Launch a virtual machine and SSH into it.

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Introduction and Overview

In this Proof of Concept (PoC), we will explore how to deploy a static website using GitHub Pages. GitHub Pages is a free hosting service provided by GitHub that allows users to serve static content directly from a Git repository. This process enables developers to host portfolios, documentation, and other web-based projects without needing a dedicated server. By following the steps outlined, we will successfully set up a GitHub repository, push local website files, and enable GitHub Pages to host the static website.

Objective

The goal of this project is to:

1. Create a GitHub repository for hosting a static website.
2. Push local website files to the repository.
3. Configure GitHub Pages to serve the website.
4. Obtain and verify the live URL of the hosted website.
5. Maintain and update the website when needed.

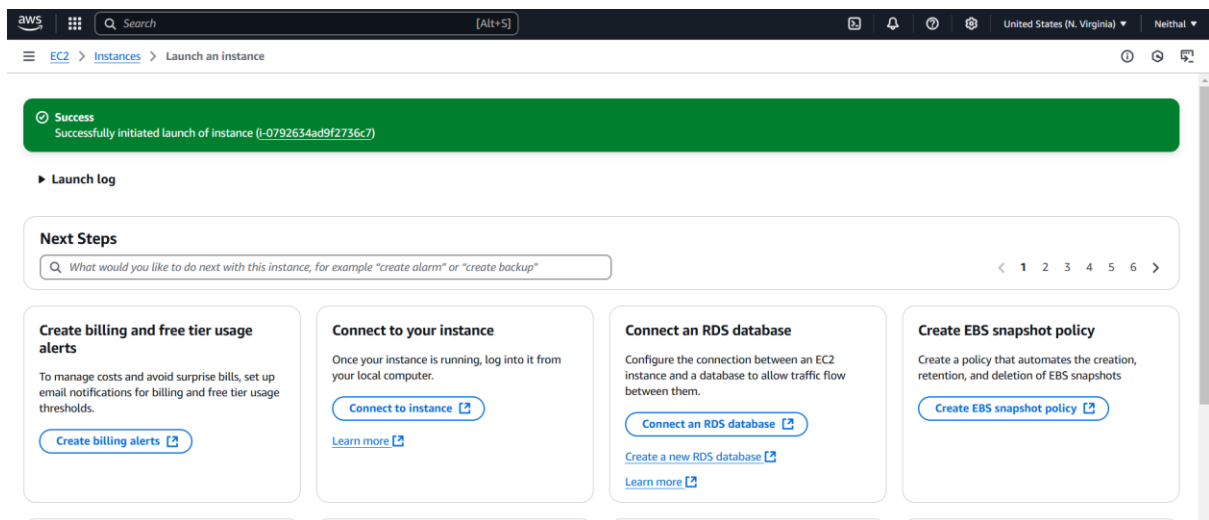
Importance of Deploying Websites Using GitHub Pages

Deploying static websites via GitHub Pages offers numerous advantages, including:

- **Free Hosting:** Eliminates the need for paid hosting services.
- **Easy Maintenance:** Updates are as simple as pushing new changes to the repository.
- **Version Control Integration:** Enables tracking of website modifications using Git.
- **Custom Domain Support:** Allows users to configure their own domain names.
- **Secure and Reliable:** Ensures a stable and secure hosting environment.

Steps for Execution

Step 1: Create a GitHub Repository First, log in to GitHub and create a new public repository. Ensure the repository name is relevant to your project. Initialize the repository with a README file, which will serve as the initial commit. Copy the repository URL for future reference.



Step 2: Prepare and Push Website Files On your local system, create or navigate to the folder containing your static website files (HTML, CSS, and JavaScript). Initialize a Git repository and add a remote pointing to the GitHub repository created in Step 1. Commit the website files and push them to the GitHub repository.

```
ubuntu@ip-172-31-87-198: ~  
Expanded Security Maintenance for Applications is not enabled.  
0 updates can be applied immediately.  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
ubuntu@ip-172-31-87-198:~$ |
```

Step 3: Enable GitHub Pages Navigate to the repository settings on GitHub and locate the "Pages" section under "Build and Deployment." Select "Deploy from a branch" and choose the appropriate branch (usually main or gh-pages). Save the settings to enable GitHub Pages for the repository.

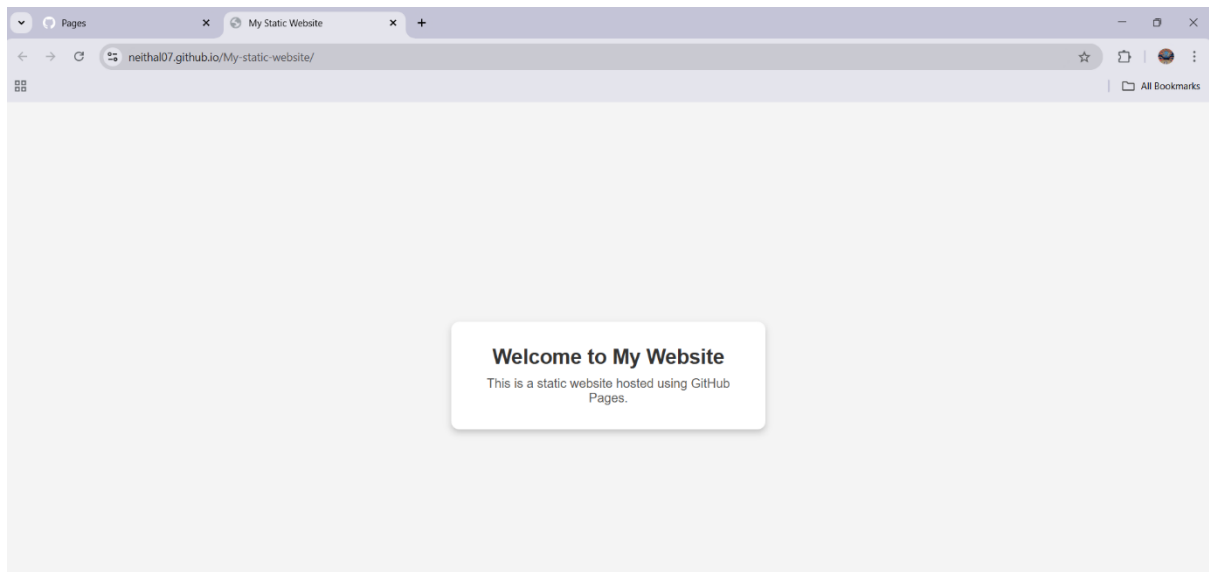
```
ubuntu@ip-172-31-87-198:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.1 LTS
Release:        24.04
Codename:       noble
ubuntu@ip-172-31-87-198:~$
```

Step 4: Obtain and Verify the Website URL Once GitHub Pages is enabled, GitHub will generate a URL for your website, typically in the format `https://yourusername.github.io/repository-name/`. Open this URL in a browser to verify that your website is successfully deployed.

```
ubuntu@ip-172-31-87-198: ~
See "man sudo_root" for details.

ubuntu@ip-172-31-87-198:~$ chmod 400 my-key.pem
chmod: cannot access 'my-key.pem': No such file or directory
ubuntu@ip-172-31-87-198:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.1 LTS
Release:        24.04
Codename:       noble
ubuntu@ip-172-31-87-198:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           957Mi       310Mi       422Mi       888Ki       376Mi       647Mi
Swap:              0B           0B           0B
ubuntu@ip-172-31-87-198:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.8G  1.7G  5.1G  26% /
tmpfs            479M   0    479M   0% /dev/shm
tmpfs            192M  876K  191M   1% /run
tmpfs            5.0M   0    5.0M   0% /run/lock
/dev/xvda16      881M   76M  744M  10% /boot
/dev/xvda15      105M   6.1M   99M   6% /boot/efi
tmpfs            96M   12K   96M   1% /run/user/1000
ubuntu@ip-172-31-87-198:~$ |
```

Step 5: Maintain and Update the Website To update the website, make changes to the local files, commit the modifications, and push them to the repository. GitHub Pages will automatically update the hosted version with the latest changes.



Conclusion By following these steps, we successfully deployed a static website using GitHub Pages. This method provides an efficient and free solution for hosting personal projects, portfolios, and documentation sites.

What I Learned from This Process

- **GitHub Pages Hosting:** Understanding how to host static websites without a server.
- **Git Workflow:** Managing a project repository using Git commands.
- **Version Control:** Tracking website modifications through commits.
- **Website Maintenance:** Keeping a hosted website up to date with new changes.
- **GitHub Settings:** Configuring repository settings to enable GitHub Pages hosting.